

SHUBHI DIXIT

Aspiring Software Engineer | First-Year B.Tech CSE Student | DSA • Web Dev • Machine Learning • AI-ML

[Gmail](#) ♦ [LinkedIn](#) ♦ [LeetCode](#) ♦ [Codeforces](#) ♦ [Github](#)

SUMMARY

Computer Science undergraduate at Delhi Technological University with a strong foundation in Data Structures & Algorithms, full-stack web development, and distributed systems. Deeply passionate about Artificial Intelligence, Machine Learning, and building scalable software solutions. Proficient in C++, Python, JavaScript, and the MERN stack with hands-on experience in multi-agent AI systems and optimization algorithms.

EDUCATION

B.Tech (Computer Science Engineering)- Delhi Technological University 2025-2029

CGPA: 6.5

Relevant Coursework: Data Structures, Machine Learning, MERN

Class X (CBSE) 95.8% 2021-2022

Class XII (CBSE) 89.8% 2024-2025

SKILLS

Languages	C++, Python, JavaScript
Technical Skills	Data Structures and Algorithms, Web Development (MERN), Machine Learning, Operating System, LLMs and RAG, OOP.
Tools	NodeJS, Express JS, ReactJS, MongoDB, SQLite, Git and GitHub, Canvas API.

RANKS AND ACHIEVEMENTS

JEE Main: 99.08 percentile

Qualified JEE Advanced

1st Place, *Guessapalooza* | IEEE DTU INVICTUS Annual Technical Fest | ₹3,000 Cash Prize

2nd Place, State-Level Mental Mathematics Competition (Class XII) | ₹10,500 Cash Prize

Rank 12, MVPP Delhi State Scholarship Examination (Class IX) among 20,000+ students | ₹5,000 Scholarship

PROJECTS

NyayaBot | *Python, TypeScript, React, Gemma 4, Ollama, ChromaDB, SQLite*

- Built a **100% on-device, offline legal-assistance engine** on **Gemma 4**, using Per-Layer Embeddings to run inference under **6GB VRAM** with zero cloud dependency, ensuring full query privacy.
- Designed a **hierarchical RAG pipeline** over **ChromaDB** that retrieves the relevant Act before narrowing to the exact Section, paired with a custom **guardrail layer (ShieldAI)** enforcing citation verification and output safety checks to prevent hallucinated legal answers.
- Engineered a **ReAct-loop workflow orchestrator** coordinating 8 tools (statutory search, document drafting, citation verification) over a **13-table SQLite schema** with **SQL-level compare-and-swap concurrency** and idempotency-key validation for atomic, conflict-safe case writes.

SkillForge | *Python, FastAPI, LangGraph, OpenCV, Docker, Git* | (In Progress)

- Building a **modular Agentic AI runtime** that orchestrates specialized AI agents, memory, and external tools to autonomously execute complex workflows from natural-language instructions.
- Designing a **multi-agent architecture** comprising **Planning, Vision, Memory, and Execution** agents that collaborate through a shared workflow and event-driven execution framework for modular task orchestration.
- Developing an extensible runtime integrating **computer vision, tool execution, browser automation, and simulation-ready adapters**, with a long-term vision of supporting **robotics and sim-to-real deployment**.

ChronoSync | *ReactJS, Node.js, Express.js, MongoDB, Genetic Algorithm, Greedy Algorithm* | (SIH 2025)

- Built a **prototype web-based timetable scheduling system** for Indian universities, supporting **multi-department scheduling, shared classroom/lab allocation, and faculty preference management**.
- Designed a **hybrid Genetic Algorithm + Greedy Optimization framework** to satisfy **hard constraints** (teacher, room, and batch conflicts) while optimizing **soft constraints** including faculty preferences and workload distribution.
- Developed a **role-based web platform** with Admin, Teacher, and Student portals, featuring **automated timetable generation, conflict detection, schedule regeneration**, and seamless integration with existing university **ERP systems**.

EXTRA-CURRICULAR ACTIVITIES

- Volunteered with NSS, mentoring underprivileged children and conducting community outreach to promote government educational initiatives.
- Participated in interschool and state-level competitions in Poetry, Debate, and Arts, strengthening communication, creativity, and public speaking skills.
- Completed a 2-month Summer Program at the British Council, enhancing spoken English, creative writing, presentation, teamwork, and leadership skills.